

Probability & Statistics

Tutorials 1-5: Questions and Detailed Exam-Style Solutions

Compiled from the five uploaded tutorial/practice files. Solutions are written in an exam-friendly, step-by-step format.

Important source note: A few questions in the uploaded files have incomplete/garbled table data. Those are clearly marked rather than being silently reconstructed.

Topics covered: probability laws, conditional probability, Bayes' theorem, diagnostic testing, blood groups, independence, Chi-square, uniform, normal and gamma distributions, mean, median, mode, quartiles, variance, standard deviation, grouped data, histograms and frequency distributions.

Tutorial 1 — Probability

The uploaded Tutorial 1 contains Questions 1-8 on probability and conditional probability.

Q1(i). Prove $P(A) = P(A \cap B) + P(A \cap B')$

Concept: Whenever A occurs, either B occurs or B does not occur.

Therefore, A can be divided into two mutually exclusive parts:

$$A = (A \cap B) \cup (A \cap B')$$

Since the two parts do not overlap, their probabilities add:

$$P(A) = P(A \cap B) + P(A \cap B')$$

Final answer: $P(A) = P(A \cap B) + P(A \cap B')$

Q1(ii). Prove $P(A \cup B) = P(A \cap B) + P(A \cap B') + P(A' \cap B)$

The event $A \cup B$ consists of three mutually exclusive cases:

1. A and B occur: $A \cap B$
2. A occurs but B does not: $A \cap B'$
3. B occurs but A does not: $A' \cap B$

Hence:

$$A \cup B = (A \cap B) \cup (A \cap B') \cup (A' \cap B)$$

Adding the probabilities of these mutually exclusive parts gives the required result.

Final answer: $P(A \cup B) = P(A \cap B) + P(A \cap B') + P(A' \cap B)$

Q2. At least one event has probability p; exactly one has probability q

Given $P(A \cup B) = p$ and $P(\text{exactly one of } A, B) = q$.

We have $P(A \cup B) = P(A) + P(B) - P(A \cap B) = p$.

Also, $q = P(A \cap B') + P(A' \cap B)$. Since $P(A) + P(B) = 2P(A \cap B) + q$, combining with the first equation gives:

$$P(A \cap B) = p - q.$$

Therefore $P(A) + P(B) = p + (p - q) = 2p - q$.

Using $P(A') = 1 - P(A)$ and $P(B') = 1 - P(B)$:

$$P(A') + P(B') = 2 - [P(A) + P(B)] = 2 - (2p - q).$$

Final answer: $P(A') + P(B') = 2 - 2p + q$

Q3. Two dice are thrown

One die is blue and one is grey, so outcomes are ordered pairs (blue, grey). There are $6 \times 6 = 36$ equally likely outcomes.

All outcomes: (1,1), (1,2), ..., (1,6); (2,1), ..., (2,6); (3,1), ..., (3,6); (4,1), ..., (4,6); (5,1), ..., (5,6); (6,1), ..., (6,6).

(i) Sum = 8: favorable outcomes are (2,6), (3,5), (4,4), (5,3), (6,2): 5 outcomes.

Final answer: $P(\text{sum} = 8) = 5/36$

(ii) Sum = 13: the maximum sum is $6+6=12$, so sum 13 is impossible.

Final answer: $P(\text{sum} = 13) = 0$

(iii) $\text{Sum} \leq 12$: every possible outcome has sum at most 12.

Final answer: $P(\text{sum} \leq 12) = 1$

Q4. Birthday probability

Given $P(\text{two students do not have the same birthday}) = 0.992$.

The required event is its complement:

$P(\text{same birthday}) = 1 - 0.992 = 0.008$.

Final answer: 0.008, or 0.8%

Q5. Marbles

Colour	Number
Blue	3
White	2
Red	4
Total	9

Use $P(\text{event}) = \text{favorable outcomes} / \text{total outcomes}$.

(i) White: $2/9$. (ii) Blue: $3/9 = 1/3$. (iii) Red: $4/9$.

Final answer: (i) $2/9$; (ii) $1/3$; (iii) $4/9$

Q6. Red and black balls

There are $3 \text{ red} + 5 \text{ black} = 8 \text{ balls}$.

(i) $P(\text{red}) = 3/8$. (ii) $P(\text{not red}) = P(\text{black}) = 5/8$.

Final answer: (i) $3/8$; (ii) $5/8$

Q7. At least one Group AB subject among 20

Given $P(\text{AB})=0.0526$. It is easier to calculate the complement: nobody has AB blood.

$P(\text{not AB for one subject}) = 1 - 0.0526 = 0.9474$.

Assuming independent selections: $P(\text{no AB in 20}) = (0.9474)^{20}$.

Therefore $P(\text{at least one AB}) = 1 - (0.9474)^{20} \approx 0.6606$.

Final answer: Approximately 0.661, or 66.1%. This is greater than 50%, but it is not usually high enough to claim strong certainty.

Q8. Cancer test — probability of cancer given a positive test

Given prevalence $P(C)=0.01$, false-positive rate $P(+|C')=0.10$, and true-positive rate $P(+|C)=0.80$.

Bayes' theorem:

$P(C|+) = [P(+|C)P(C)] / [P(+|C)P(C) + P(+|C')P(C')]$.

$P(C')=0.99$.

$P(C|+) = (0.80 \times 0.01) / [(0.80 \times 0.01) + (0.10 \times 0.99)] = 0.008 / 0.107 \approx 0.0748$.

Final answer: $P(C|+) \approx 0.0748 = 7.48\%$

Notation note: The final line of the uploaded question mentions a negative test while the preceding wording asks for a positive test. If the intended question is $P(C|-)$, then $P(-|C)=0.20$ and $P(-|C')=0.90$, giving $P(C|-) \approx 0.00224 = 0.224\%$.

Tutorial 2 — Conditional Probability, Bayes and Blood Groups

Q1. Cancer given a positive test

Given $P(C)=0.01$, $P(C')=0.99$, $P(+|C)=0.80$, $P(+|C')=0.10$.

Bayes' theorem:

$$P(C|+) = (0.80 \times 0.01) / [(0.80 \times 0.01) + (0.10 \times 0.99)] = 0.008 / 0.107.$$

Final answer: $P(C|+) \approx 0.0748 = 7.48\%$

Q2(a). Positive test given that the subject uses drugs

From the 555-subject table in the uploaded tutorial, 50 subjects use drugs and 45 of those test positive.

$$P(+|D) = 45/50 = 0.90.$$

Final answer: $0.90 = 90\%$

Q2(b). Subject uses drugs given a positive test

There are 70 positive test results, of which 45 are from actual drug users.

$$P(D|+) = 45/70 = 9/14 \approx 0.6429.$$

Final answer: $0.6429 = 64.29\%$

Exam warning: $P(+|D)$ and $P(D|+)$ are different conditional probabilities. Do not interchange them.

Q3. Blood groups and Rh types

	O	A	B	AB	Total
Rh+	59	53	12	6	130
Rh-	9	8	3	2	22
Total	68	61	15	8	152

(a) Not Group A: A total = 61, so not A = $152 - 61 = 91$.

Final answer: $91/152 \approx 0.5987 = 59.87\%$

(b) Not Rh+: Rh- total = 22.

Final answer: $22/152 = 11/76 \approx 0.1447 = 14.47\%$

(c) Group A or Rh+: $P(A \cup \text{Rh}+) = [61 + 130 - 53] / 152 = 138 / 152$.

Final answer: $138/152 \approx 0.9079 = 90.79\%$. The events are not disjoint because 53 subjects are both A and Rh+.

(d) Rh- or Group AB: $[22 + 8 - 2] / 152 = 28 / 152$.

Final answer: $28/152 = 7/38 \approx 0.1842 = 18.42\%$. The events are not disjoint because 2 subjects are both Rh- and AB.

(e) Both Group B: There are 15 Group B subjects.

With replacement: $(15/152)(15/152) \approx 0.00974$; selections are independent.

Without replacement: $(15/152)(14/151) \approx 0.00915$; selections are not independent.

Final answer: With replacement ≈ 0.00974 (independent); without replacement ≈ 0.00915 (not independent).

(f) Both Rh-: With replacement: $(22/152)^2 \approx 0.02095$. Without replacement: $(22/152)(21/151) \approx 0.02013$.

Final answer: With replacement ≈ 0.02095 (independent); without replacement ≈ 0.02013 (not independent).

(g) Both Rh+: With replacement: $(130/152)^2 \approx 0.7315$. Without replacement: $(130/152)(129/151) \approx 0.7307$.

Final answer: With replacement ≈ 0.7315 (independent); without replacement ≈ 0.7307 (not independent).

(h) Both Group AB: With replacement: $(8/152)^2 \approx 0.00277$. Without replacement: $(8/152)(7/151) \approx 0.00244$.

Final answer: With replacement ≈ 0.00277 (independent); without replacement ≈ 0.00244 (not independent).

(i) Group A or Group B or Rh-: Count = $61+15+22-8-3 = 87$.

Final answer: $87/152 \approx 0.5724 = 57.24\%$

(j) Group O or Group AB or Rh+: Count = $68+8+130-59-6 = 141$.

Final answer: $141/152 \approx 0.9276 = 92.76\%$

(k) Three different people, all Group A:

$P = (61/152)(60/151)(59/150) \approx 0.06272$.

Final answer: $\approx 0.06272 = 6.27\%$

(l) Three different people, all Rh-:

$P = (22/152)(21/151)(20/150) \approx 0.002684$.

Final answer: $\approx 0.002684 = 0.268\%$

Tutorial 3 — Probability Distributions

Q1. Chi-square distribution with 6 degrees of freedom

For a Chi-square random variable with ν degrees of freedom:

Mean = ν and Variance = 2ν .

Here $\nu=6$.

Final answer: Mean = 6; Variance = 12

Q2. Uniform density

Let $X \sim \text{Uniform}(\alpha, \beta)$, so $f(x) = 1/(\beta - \alpha)$ for $\alpha < x < \beta$.

Let $c = \alpha + p(\beta - \alpha)$. Then:

$$P[X < c] = \int_{\alpha}^c \frac{1}{(\beta - \alpha)} dx = (c - \alpha)/(\beta - \alpha).$$

Substituting c gives $[p(\beta - \alpha)]/(\beta - \alpha) = p$.

Final answer: $P[X < \alpha + p(\beta - \alpha)] = p$

Q3(a). $P(Z < 1.33)$

From the standard normal table, $\Phi(1.33) = 0.9082$.

Final answer: 0.9082

Q3(b). $P(Z \geq -0.79)$

$P(Z \geq -0.79) = 1 - P(Z < -0.79)$. From the table $P(Z < -0.79) = 0.2148$.

Final answer: 0.7852

Q3(c). $P(0.55 < Z < 1.22)$

$$= \Phi(1.22) - \Phi(0.55) = 0.8888 - 0.7088.$$

Final answer: 0.1800

Q3(d). $P(-1.90 \leq Z \leq 0.44)$

$$= \Phi(0.44) - \Phi(-1.90) = 0.6700 - 0.0287.$$

Final answer: 0.6413

Q4(a). Gamma distribution, $\alpha=2$, $\beta=3$

Using the shape-scale convention for $\text{Gamma}(\alpha, \beta)$, for integer α :

$$P(X > x) = e^{-(x/\beta)} \sum_{k=0}^{\alpha-1} (x/\beta)^k / k!.$$

For $x=4$, $\alpha=2$, $\beta=3$:

$$P(X > 4) = e^{-(4/3)} [1 + 4/3] \approx 0.6151.$$

Final answer: ≈ 0.6151

Parameterization note: Some textbooks use β as a rate instead of a scale. If your class uses the rate convention, the numerical result changes. This solution uses β as the scale parameter.

Q4(b). Gamma distribution, $\alpha=3$, $\beta=4$

$$P(X > 4) = e^{-(1)} [1 + 1 + 1/2] = 2.5/e \approx 0.9197.$$

Final answer: ≈ 0.9197

Q5(a). Area between 0 and $z = 0.4726$

Area to the left of $z = 0.5 + 0.4726 = 0.9726$. From the Z table, $z \approx 1.92$.

Final answer: $z \approx 1.92$

Q5(b). Area to the left of $z = 0.9868$

Directly look up 0.9868 in the standard normal table.

Final answer: $z \approx 2.22$

Q5(c). Area to the right of $z = 0.1314$

Left area = $1 - 0.1314 = 0.8686$. From the table, $z \approx 1.12$.

Final answer: $z \approx 1.12$

Q5(d). Area between $-z$ and $z = 0.8502$

Total tail area = $1 - 0.8502 = 0.1498$. Each tail = 0.0749. Thus left area to $z = 0.9251$, giving $z \approx 1.44$.

Final answer: $z \approx 1.44$

Q6(a). Normal variable: $P(X > 5)$

Given $\mu = 10$ and $\sigma^2 = 36$, so $\sigma = 6$.

$Z = (5 - 10)/6 = -0.8333$. Therefore $P(X > 5) = P(Z > -0.8333) \approx 0.7977$.

Final answer: ≈ 0.7977

Q6(b). $P(4 < X < 16)$

For $X = 4$, $z = (4 - 10)/6 = -1$. For $X = 16$, $z = 1$.

$P(-1 < Z < 1) \approx 0.6827$.

Final answer: ≈ 0.6827

Q6(c). $P(X < 8)$

$z = (8 - 10)/6 = -0.3333$. Hence $P(Z < -0.3333) \approx 0.3694$.

Final answer: ≈ 0.3694

Q6(d). $P(X < 20)$

$z = (20 - 10)/6 = 1.6667$. Hence $P(Z < 1.67) \approx 0.9522$.

Final answer: ≈ 0.9522

Q6(e). $P(X > 16)$

$z = (16 - 10)/6 = 1$. Therefore $P(Z > 1) = 1 - 0.8413 = 0.1587$.

Final answer: 0.1587

Tutorial 4 — Descriptive Statistics and Data Presentation

Q1. Three coins tossed 30 times

Observed numbers of heads: 0,1,2,2,1,2,3,1,3,0; 1,3,1,1,2,2,0,1,2,1; 3,0,0,1,1,2,3,2,2,0.

Number of heads	Frequency
0	6
1	10
2	9
3	5
Total	30

For a bar plot, put number of heads on the x-axis and frequency on the y-axis. Bar heights are 6, 10, 9 and 5 respectively.

Q2. Car battery lifetimes — grouped frequency

Class width = 0.5 starting at 2.0–2.5.

Class interval	Frequency	Midpoint
2.0–2.5	2	2.25
2.5–3.0	6	2.75
3.0–3.5	14	3.25
3.5–4.0	11	3.75
4.0–4.5	4	4.25
4.5–5.0	3	4.75
Total	40	

Histogram: use the class intervals on the x-axis and frequencies on the y-axis; the bars touch.

Frequency polygon: plot midpoint-frequency points (2.25,2), (2.75,6), (3.25,14), (3.75,11), (4.25,4), (4.75,3) and join them.

Q3. Mean and median salary of employees

Source limitation: The salary/frequency table in the uploaded document is extracted with missing/garbled rows. The available text shows salary/frequency fragments but not a complete reliable table. Therefore a numerical answer should not be invented.

Correct method when the complete table is available:

Mean = $\Sigma fx / \Sigma f$.

Median: calculate cumulative frequencies, locate the middle observation(s), and use the appropriate discrete/grouped median method.

Final answer: Numerical result cannot be determined reliably from the uploaded table as extracted.

Q4. Mode of 14, 25, 14, 28, 18, 17, 18, 14, 23, 22, 14, 18

Frequency of 14 = 4; frequency of 18 = 3; all other values occur fewer times.

Final answer: Mode = 14

Q5. Children of different age groups

Age	Frequency
1-2	5
2-3	3
3-5	6
5-7	12
7-10	9
10-15	10
15-17	4

(a) Histogram: Because class widths are unequal, use frequency density = frequency/class width.

(b) Mean: Use class midpoints and $\Sigma fx/\Sigma f$. The uploaded tutorial gives mean = 7.68.

Final answer: Mean = 7.68

(c) Median: Total N=49, so the median position is 24.5. Locate the median class using cumulative frequency and apply the grouped median formula.

Final answer: Median = 6.75

Q6. Average weight from average height

Average height = 1.4 m and weight $W=40H$.

Taking averages: $W=40H=40(1.4)=56$.

Final answer: Average weight = 56

Q7. Mean of y

Source limitation: The uploaded document says the mean of x is 30 but the equation defining y in terms of x is missing from the extracted question.

If the missing relation is $y=ax+b$, then $\bar{y}=a\bar{x}+b=30a+b$.

Final answer: A numerical value cannot be determined without the missing relationship between x and y.

Q8. Winning golf scores, 2004-2013

Scores: 280, 278, 272, 276, 281, 279, 276, 281, 289, 280.

Sum = 2792 and $n=10$.

Sample mean = $2792/10 = 279.2$.

Final answer: Sample mean = 279.2

Sorted data: 272, 276, 276, 278, 279, 280, 280, 281, 281, 289. Median = $(279+280)/2$.

Final answer: Median = 279.5

Q9. Clay shrinkage

There are 50 observations. The uploaded tutorial reports mean = 18.97.

Final answer: Mean $\approx$ 18.97 (as supplied in the tutorial)

A histogram can be constructed by choosing sensible class intervals and plotting frequency against shrinkage percentage. For a sample standard deviation, use $s = \sqrt{[\Sigma(x-\bar{x})^2/(n-1)]}$.

Q10. Pedestrians killed in fatal road accidents — male and female means

Use each age-group midpoint x and frequency f , then $\bar{x} \approx \Sigma fx / \Sigma f$.

The uploaded tutorial gives the expected approximate means:

Final answer: Male mean = 40.90; Female mean = 38.88

Tutorial 5 — Statistics Practice Sheet

Q1. Coal ash content

Data: 9.2, 14.1, 9.8, 12.4, 16.0, 12.6, 22.7, 18.9, 21.0, 14.5, 20.4, 16.9.

There are $n=12$ observations. Sum = 188.5.

Sample mean = $188.5/12 = 15.70833$.

Final answer: Sample mean = 15.70833

Sample standard deviation uses $s = \sqrt{[\sum(x-\bar{x})^2]/(n-1)}$. The supplied answer is 4.39534.

Final answer: Sample standard deviation = 4.39534

Q2. Five observations: mean 104, sample variance 16

Let the missing values be x and y . Since $n=5$ and $\bar{x}=104$, total sum = $5 \times 104 = 520$.

Known values sum = $102+100+105=307$, so $x+y=213$.

Using $s^2 = 16$: $\sum(x_i-104)^2 = (n-1)s^2 = 4 \times 16 = 64$.

Equivalently, $\sum x_i^2 = (n-1)s^2 + n\bar{x}^2 = 4(16) + 5(104^2) = 54144$.

Known squares = $102^2+100^2+105^2=31429$, so $x^2+y^2=22715$.

$(x+y)^2=x^2+y^2+2xy$ gives $213^2=22715+2xy$, hence $xy=11327$.

Thus $t^2-213t+11327=0$. Solving gives $t=(213 \pm \sqrt{61})/2$.

Final answer: The two missing values are approximately 102.595 and 110.405.

Q3. Transistor lifetimes

There are 40 lifetimes. The uploaded tutorial gives mean = 127.425 and lists the modes.

Final answer: Mean = 127.425; Median = 127.5

The dataset is multimodal. Values occurring most often are 108, 118, 121, 126, 130, 132, 134, 136 and 140.

Final answer: Modes = 108, 118, 121, 126, 130, 132, 134, 136, 140

Using the median-of-halves convention for quartiles:

Final answer: Q1 = 118.5; Q2 = 127.5; Q3 = 135.5

Quartile conventions differ across textbooks/software. If your instructor specifies a different convention, follow that convention.

Q4. Worldwide fatal airline accidents, 1997-2005

Data: 25, 20, 21, 18, 13, 13, 7, 9, 18.

$n=9$ and mean = 16.

Squared deviations from 16 sum to 278.

Sample variance = $278/(9-1) = 34.75$.

Final answer: Sample variance = 34.75

Q5. When is mean appropriate?

(i) Mean appropriate: Heights of students in a relatively homogeneous class with no extreme outliers.

Final answer: Example: student heights in a fairly symmetric dataset.

(ii) Median appropriate: Salaries or house prices with extreme outliers. A few very large values can pull the mean away from the typical observation.

Final answer: Example: income data with a few extremely high salaries.

Q6. Grouped frequency data

Class	f	Midpoint
0-20	5	10
20-40	7	30
40-60	6	50
60-80	9	70
80-100	10	90
100-120	12	110
Total	49	

Mean = $\Sigma fx / \Sigma f = 3410 / 49 \approx 69.59$.

Final answer: Mean ≈ 69.59

Median position = $N/2 = 24.5$. The median class is 60-80. Using $L=60$, $CF=18$, $f=9$, $h=20$:

Median = $60 + [(24.5 - 18) / 9] \times 20 \approx 74.44$.

Final answer: Median ≈ 74.44

The modal class is 100-120. Using the grouped mode formula with $f_1=12$, $f_0=10$ and $f_2=0$ gives approximately 102.86.

Final answer: Modal class = 100-120; grouped mode ≈ 102.86

Using population-style grouped variance with class midpoints gives approximately 1118.20; standard deviation ≈ 33.44 .

Final answer: Variance ≈ 1118.20 ; standard deviation ≈ 33.44

If the question explicitly asks for sample variance, use $N-1$ instead of N ; the result is approximately 1141.50 and $s \approx 33.79$.

Q7. First 99 values have mean 120; final 99 have mean 100

There are 198 observations, with equal group sizes of 99.

Overall mean = $[99(120) + 99(100)] / 198 = 110$.

Final answer: Overall sample mean = 110

For the median, the two group means do not tell us the ordering of the observations, so the overall median cannot be determined.

Final answer: Overall median cannot be determined from the given information.

For the mode, the frequencies of individual values are not provided, so the overall mode cannot be determined.

Final answer: Overall mode cannot be determined from the given information.

Q8. Pedestrians crossing a street — male/female means and quartiles

The table gives age intervals and male/female frequencies. Use class midpoints for the approximate means.

For grouped data: $\bar{x} \approx \Sigma fx / \Sigma f$.

Final answer: Using the table: male mean ≈ 40.90 ; female mean ≈ 40.99 .

For grouped quartiles use $Q = L + [(position - CF) / f]h$.

Final answer: Using the common grouped-data interpolation: male $Q1 \approx 8.35$, $Q2 \approx 46.92$, $Q3 \approx 67.35$; female $Q1 \approx 8.54$, $Q2 \approx 48.25$, $Q3 \approx 66.63$.

The tutorial's Tutorial 5 table differs from the fatal-accident table in Tutorial 4, so the tables should not be mixed.

Final Exam Formula Sheet

Concept	Formula
Complement	$P(A') = 1 - P(A)$
Addition theorem	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional probability	$P(A B) = P(A \cap B) / P(B)$
Bayes theorem	$P(A B) = P(B A)P(A) / P(B)$
Independent events	$P(A \cap B) = P(A)P(B)$
Mean	$\bar{x} = \Sigma x / n$
Frequency mean	$\bar{x} = \Sigma fx / \Sigma f$
Sample variance	$s^2 = \Sigma (x - \bar{x})^2 / (n - 1)$
Sample SD	$s = \sqrt{s^2}$
Z-score	$Z = (X - \mu) / \sigma$
Grouped median	$\text{Median} = L + [(N/2 - CF) / f]h$
Grouped mode	$\text{Mode} = L + [(f_1 - f_0) / (2f_1 - f_0 - f_2)]h$
Chi-square mean	ν
Chi-square variance	2ν
Uniform density	$f(x) = 1 / (\beta - \alpha)$

High-Priority Exam Reminders

1. Do not confuse $P(+|C)$ with $P(C|+)$.
2. For sample variance and sample standard deviation, the denominator is $n-1$.
3. For a normal variable, convert X to Z using $Z = (X - \mu) / \sigma$.
4. If σ^2 is given, take the square root to obtain σ .
5. With replacement generally leaves probabilities unchanged and supports independence; without replacement changes the composition and generally creates dependence.
6. For grouped data, use class midpoints for the mean and cumulative frequencies for the median/quartiles.
7. For unequal class widths in a histogram, use frequency density rather than raw frequency as the bar height.
8. If source data are incomplete, state the limitation rather than inventing missing values.